

Zero & Negative Exponents — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (c) **1/9**. $3^{-2} = 1/3^2 = 1/9$.

Q2 (a) **16**. $1/4^{-2} = 4^2 = 16$.

Q3 **1**. Any non-zero number to the power 0 is 1.

Q4 (c) **3¹¹**. $5 - (-6) = 11$.

Q5 **True**. A negative power flips the fraction.

Q6 (a) **Both true, R explains A**. $2^{-3} = 1/2^3 = 1/8$.

Short answer

Q7 **2⁻⁸**. $16 = 2^4$, so $16^{-2} = (2^4)^{-2} = 2^{-8}$.

Q8 (i) **25/4** (ii) **16/81**. Flip then square; even power → positive.

Q9 **27/64 = (3/4)³ ; -27/64 = (-3/4)³**.

Long answer

Q10 (i) **2³ = 8** (ii) **3³ = 27** (iii) **p⁻⁷ = 1/p⁷**. ($-4+7=3$; $2-5+6=3$; $3-10=-7$.)

Q11 **(-4/3)³ = -64/27**. Flip the fraction for the negative power; odd power keeps the minus sign.

Q12 **3⁻¹ = 1/3**. $-2 + 5 - 4 = -1$.

Total: 21 marks.